求级数

$$\sum_{n=1}^{\infty}\frac{1}{n^2\binom{2n}{n}}$$

首先由于

$$\frac{n}{2}B(n,n)=\frac{1}{\binom{2n}{n}}$$

从而

$$=\int_0^1\sum_{n=1}^{\infty}\frac{1}{2n}(t-t^2)^{n-1}\mathrm{d}t$$

$$=-\int_0^1\frac{\ln(1-t+t^2)}{2t(1-t)}\mathrm{d}t$$

$$=-\frac{1}{2}\int_0^1\frac{\ln(1-t+t^2)}{t}+\frac{\ln(1-t+t^2)}{1-t}\mathrm{d}t$$

不难看出

$$\int_0^1\frac{\ln(1-t+t^2)}{t}\mathrm{d}t=\int_0^1\frac{\ln(1-t+t^2)}{1-t}\mathrm{d}t$$

从而

$$=-\int_0^1\frac{\ln(1-t+t^2)}{t}\mathrm{d}t$$

$$=\int_0^1\frac{\ln(1+t)}{t}-\frac{\ln(1+t^3)}{t}\mathrm{d}t$$

不难看出答案为

$$= \frac{\pi^2}{12}\left(1 - \frac{1}{3}\right) = \frac{\pi^2}{18}$$